Bitcoin Market Sentiment and Trader Performance Analysis

Executive Summary

This study investigates the relationship between Bitcoin market sentiment and trader performance using 211,224 Hyperliquid trades combined with the Bitcoin Fear & Greed Index.

The analysis reveals a measurable relationship between sentiment and profitability. Extreme Greed periods produced the highest average profit per trade and the highest win rate, while Fear periods generated the highest total profits and the largest average position sizes.

The findings suggest that market sentiment influences trader behavior, risk allocation, and profitability, making sentiment a potentially valuable input for trading strategies.

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1. Introduction

Cryptocurrency markets are highly influenced by investor psychology. The Fear & Greed Index is commonly used to quantify market sentiment and capture periods of optimism and pessimism.

This project examines whether trader performance changes across different sentiment regimes and whether sentiment-aware trading strategies can improve decision making.

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2. Data Overview

Hyperliquid Historical Trading Dataset

- Total Trades: 211,224
- Features:
    - Account
    - Coin
    - Execution Price
    - Trade Size

- Direction
- Closed PnL
- Timestamp

Bitcoin Fear & Greed Dataset

- Daily Sentiment Score
- Daily Sentiment Classification:
  - Extreme Fear
  - Fear
  - Neutral
  - Greed
  - Extreme Greed

After preprocessing, both datasets were merged using trading date to associate every trade with prevailing market sentiment.

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3. Methodology

1. Data Cleaning and Validation
2. Timestamp Standardization
3. Date-Based Dataset Merging
4. Exploratory Data Analysis
5. Profitability Analysis
6. Win Rate Analysis
7. Position Size Analysis
8. Coin-Level Analysis
9. Trader-Level Analysis

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4. Key Findings

Finding 1: Extreme Greed Produces Highest Average Profitability

Sentiment	Average PnL

Extreme Greed	67.89
Fear	54.29
Greed	42.74
Extreme Fear	34.54
Neutral	34.31

Interpretation:

Bullish market environments create favorable conditions for profitable trading. Traders generated nearly double the average profit during Extreme Greed compared with Neutral sentiment periods.

Finding 2: Fear Generates Highest Aggregate Profit

Sentiment	Total PnL
Fear	3.36 Million
Extreme Greed	2.72 Million
Greed	2.15 Million

Interpretation:

Although average profitability peaks during Extreme Greed, Fear periods create a larger volume of profitable opportunities.

Finding 3: Traders Deploy More Capital During Fear

Sentiment	Average Position Size (USD)
Fear	7,816
Greed	5,737

Extreme Fear	5,350
Neutral	4,783
Extreme Greed	3,112

Interpretation:

Traders appear willing to commit larger amounts of capital during fearful conditions, potentially attempting to accumulate assets at discounted prices.

Finding 4: Win Rates Improve During Extreme Greed

Sentiment	Win Rate
Extreme Greed	46.49%
Fear	42.08%
Neutral	39.70%
Greed	38.48%
Extreme Fear	37.06%

Interpretation:

Bullish sentiment increases the probability of successful trades.

Finding 5: Profitability Is Driven by a Small Fraction of Trades

More than 50% of all trades closed with zero realized profit.

Interpretation:

Overall profitability is driven by a relatively small subset of highly profitable trades rather than consistent profits across the majority of positions.

Finding 6: BTC, ETH, and SOL Show Strong Resilience

BTC, ETH, and SOL remained profitable across multiple sentiment regimes.

Interpretation:

Large-cap assets appear more resilient to sentiment fluctuations and may provide more stable trading opportunities.

Finding 7: Profit Generation Is Highly Concentrated

The most profitable trader generated over 2.14 million in realized profit.

Interpretation:

Trading performance is highly uneven, with a small number of participants accounting for a disproportionate share of profits.

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5. Trading Implications

The analysis suggests several practical applications:

1. Increase exposure during Extreme Greed periods where profitability and win rates are highest.
2. Monitor Fear periods for high-conviction opportunities due to larger capital deployment and higher aggregate profits.
3. Focus on highly liquid assets such as BTC, ETH, and SOL.
4. Emphasize risk management because overall profits are concentrated among relatively few successful trades.

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6. Limitations

- Leverage information was unavailable in the provided dataset.

- Results are based on historical observations and do not guarantee future performance.
- Sentiment was analyzed at the daily level rather than intraday resolution.

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7. Conclusion

The study demonstrates a clear relationship between Bitcoin market sentiment and trader performance.

Extreme Greed periods delivered the highest average profitability and win rates, while Fear periods generated the largest aggregate profits and position sizes. These findings suggest that sentiment-aware trading frameworks may improve decision making, particularly when combined with disciplined risk management and asset selection.